

Henrique Laureano

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Data Scientist | AI Engineer | Statistician | M.Sc. in Statistics and Numerical Methods in Engineering

Professional Summary

Senior AI Engineer and Statistician with a decade of experience bridging advanced mathematical theory and enterprise engineering. Expert in architecting agentic AI systems (RAG, LangGraph) and leading high-dimensional forecasting for global finance and automotive leaders (EBANX, Volvo). Graduate-trained specialist in Bayesian modeling, transforming complex data into multi-million-dollar cost optimizations and automated decision frameworks.

Experience

EBANX – Curitiba-PR, Brazil

Staff Data Scientist & AI Engineer (Finance)

[May 2025 - Present]

Led the end-to-end lifecycle - architecture, development, deployment, maintenance, and monitoring - of "Zenith", an Agentic AI system for Finance (serving analysts to C-level), automating complex reporting via robust RAG and text-to-SQL (BigQuery). Deployed Zenith to Slack and Gemini Enterprise using Flask APIs. Additionally, contributed to EBANX's proprietary Fraud Prevention ML suite, developing PySpark and Gradient Boosting pipelines via FastAPI on AWS.

Tech Stack: Languages: Python, SQL, PySpark. **Cloud:** GCP (Vertex AI, Vector Search, BigQuery) and AWS (SageMaker, EMR, Lambda, Step Functions). **AI/ML:** ADK, LangChain, PyTorch, Stan. **DevOps:** GitHub Actions, Sonar.

Senior Data Scientist (Credit Risk), Banco Bradesco – Curitiba-PR, Brazil

[Mar 2025 - Apr 2025]

Development, validation, and monitoring of credit risk models.

Tech Stack: Languages: Python, R, SAS, PySpark. **Cloud/Infrastructure:** On-premises (Hive, Hadoop), Databricks, MLFlow. **AI/ML:** PyTorch. **DevOps:** GitHub Actions, Bitbucket.

Volvo Trucks Latin America – Curitiba-PR, Brazil

Lead Data Scientist

[Jun 2023 - Feb 2025]

Led architecture and end-to-end delivery of advanced AI initiatives across LATAM. Concurrently acted as Data & AI Product Manager for one quarter, driving strategic project prioritization and executive roadmap alignment.

Senior Data Scientist (Advanced Analytics)

[Sep 2022 - May 2023]

Delivered major financial impact for LATAM HQ, generating seven-digit (9%) annual savings via improved S&A expense forecasting and six-digit monthly reductions in service costs through high-frequency telemetry modeling. Additionally, deployed Bayesian GAMMs for volume forecasting (+10% accuracy, <5% error) and optimized service contract pricing, driving a 5% sales increase and a 10% reduction in churn.

Tech Stack: Languages: Python, R, SQL, PySpark. **Cloud/Infrastructure:** On-premises (SQL, Hadoop), Databricks, MLFlow. **AI/ML:** PyTorch, Stan, INLA. **DevOps:** GitHub Actions.

Senior Statistician, Pelé Pequeno Príncipe Research Institute – Curitiba-PR, Brazil

[May 2021 – Sep 2022]

Largest exclusively pediatric hospital in Latin America. Led data visualization, sampling design, and predictive modeling for complex, infectious, and oncogenic diseases. Managed statistical workflows for epidemiological and clinical studies, as well as bioinformatics applications.

Tech Stack: Languages: Python, R, SQL. **Cloud/Infrastructure:** On-premises (SQL), Linux. **AI/ML:** PyTorch, JAX, Stan, INLA. **DevOps:** GitHub Actions.

Statistical Consultant, Freelance Statistical Consulting

[Jan 2014 – Sep 2022]

End-to-end statistical support for over 20 master's and PhD theses.

- **Diverse Field Expertise:** Agronomy, Biology, Economics, Nursing, Electrical and Chemical Engineering, Pharmacy, Education, Psychology, Chemistry, Medicine, and Zootechnics.
- **Core Focus:** Translating complex research questions into robust probabilistic frameworks, including data cleaning, hypothesis testing, and advanced modeling in **R and Python**.

Tech Stack: Languages: Python, R. **AI/ML:** PyTorch, Stan, INLA. **DevOps:** GitHub Actions.

Statistician, Telehealth Center – Hospital das Clínicas (UFMG) – Belo Horizonte-MG, Brazil

[Jul 2016 – Dec 2016]

Analyzed echocardiographic data in tropical diseases, developing predictive and statistical models to quantify risk factors for Chagas and rheumatic diseases. Special focus on competing risks and multi-state survival models. Left to pursue an MSc in Statistics at UNICAMP.

Education

Masters in Statistics and Numerical Methods in Engineering, Universidade Federal do Paraná (UFPR)

[2021]

Interrupted PhD in Applied Mathematics and Computational Sciences (AMCS),

King Abdullah University of Science and Technology (KAUST)

[2019]

Interrupted Masters in Statistics, Universidade Estadual de Campinas (UNICAMP)

[2017]

Bachelor in Statistics, Universidade Federal do Paraná (UFPR)

[2016]

Publications & Research

- **Peer-Reviewed Research:** Published **16 scientific papers** in international journals covering diverse fields such as Statistics, Medicine, and Engineering.
- **Google Scholar Profile:** [Full Publication List & Citations](#).

Licenses & Certifications

- **Certified SAFe® Product Owner / Product Manager**, Issued by Scaled Agile, Inc. (2023).
- **EF SET English Certificate (C1 Advanced)**, Issued by EF Education First. (2026).